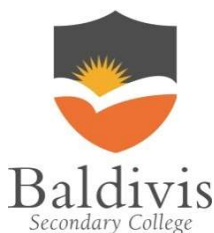


Name: _____

Date: _____



Subject - Year 8 Mathematics

Investigation 2, 2019

Topic – Circles: Round Peg in a Square Hole

$\frac{\quad}{25}$ $= \frac{\quad}{100} \%$

Total Time:	55 minutes	
Equipment:	Scientific Calculator, Interactive Notebook, Pens, Pencils	
Instructions	Show all your working out and include units of measurement to receive all available marks.	
Weighting	This is worth 5% of your grade.	

AIM: Investigate the relationship between features of circles such as circumference, area, radius and diameter. Use formulas to solve problems involving circumference and area. (ACMMG197)

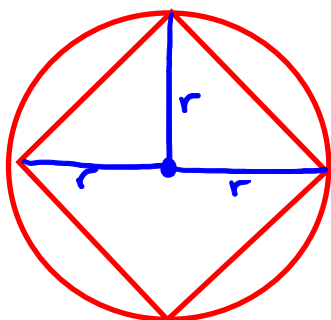
The Problem

Which is the better fit - a round peg in a square hole or a square peg in a round hole?

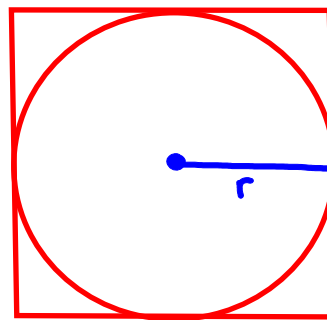
Question One

[1 mark]

Decide on and draw diagrams for a well-fitting 'round peg in a square hole' and a 'square peg in a round hole'.



Square peg in round hole



Round peg in square hole

Ensure students know how to calculate the area of the squares and circles.

Question Two

[3 marks]

Discuss what 'best fit' means and decide what quantities you will need in order to do the comparison of the pegs in holes.

Best fit - fills up the hole most completely ✓

- 1) Calculate area of the hole.
- 2) Calculate area of the peg.
- 3) Work out how much of the area of the hole is covered by the peg by using a suitable method. Students need to decide on strategy.

Question Three**[2 marks]**

Copy your diagrams from Question One and choose a value for the radius of the circle or the side length of the square in each diagram.

Diagrams copied

Value chosen for radius ✓

Value chosen for length of square ✓

Question Four *Answers will depend on values chosen* **[12 marks]**
Calculate the quantities that you need to compare. Show your working out.

Round Peg in a Square Hole	Square Peg in a Round Hole
Area of the Square Hole Formula - 1 mark Working out - 1 mark Answer with correct units - 1 mark	Area of the Round Hole Formula - 1 mark Working out - 1 mark Answer with correct units - 1 mark
Area of the Round Peg Formula - 1 mark Working out - 1 mark Answer with correct units - 1 mark	Area of the Square Peg Formula - 1 mark Working out - 1 mark Answer with correct units - 1 mark

Question Five**[2 marks]**

Which of the two situations is the better fit? Support your answer with mathematical reasoning.

Uses a suitable method to determine proportion of hole covered by the peg in each context.

Allow for fraction, percentage, or decimal comparison.

Round peg in square hole $\approx 79\%$

Square peg in round hole $\approx 64\%$

This question is open ended.

Question Six

[5 marks]

Investigate if this arrangement fits better all the time or only sometimes.

You can use other values for the radius of your circles or different lengths of your squares.

Diagrams included - 1 mark

Correct area formulas - 1 mark

Logical working out - 1 mark

Percentages/Fractions/Decimals calculated
- 1 mark

- Correct statement - 1 mark.

Students can be awarded full 5 marks even if they only use numerical values to reach a conclusion.